Reg No.:_______________ Name:__________________________

1100EET308062302

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S6 (R, S) / S6 (PT) (R) Examination June 2023 (2019 Scheme)

Course Code: EET 308

Course name: COMPREHENSIVE COURSE WORK

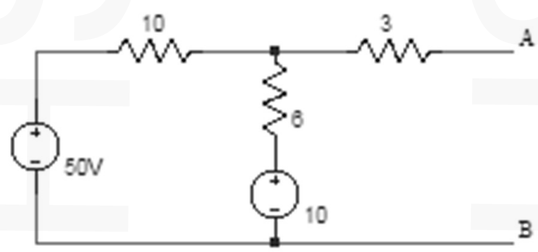
Max. Marks: 50 Duration: 1Hour

Instructions: ***(1) Each question carries one mark. No negative marks for wrong answers***
(2) Total number of questions: 50
(3) All questions are to be answered. Each question will be followed by 4 possible answers of which only ONE is correct.
(4) If more than one option is chosen, it will not be considered for valuation.

1. In Superposition theorem, while considering a source, all other current sources are?

   a) short circuited b) change its position c) open circuited d) removed from the circuit

2. Determine the equivalent Thevenin's voltage between terminals A and B in the circuit shown below.



   a) 5 b) 15 c) 25 d) 35

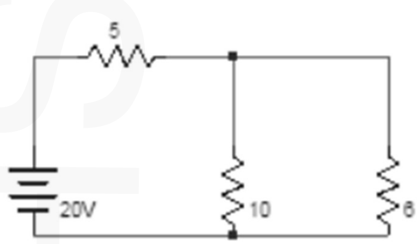
3. Kirchhoff's voltage law is based on principle of conservation of ___________

   a) energy b) momentum c) mass d) charge

4. The charge associated with a bulb rated as 20 W, 200 V and used for 10 minutes is ____

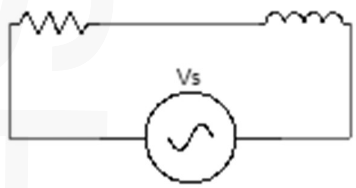
   a) 36 C b) 60 C c) 72 C d) 50 C

5. Find the voltage drop across 6Ω resistor in the circuit shown below.



a) 6.58 b) 7.58 c) 8.58 d) 9.58

6. Determine the source voltage if voltage across resistance is 3V and the voltage across inductor is 4V as shown in the figure.



a) 4 b) 5 c) 6 d) 7

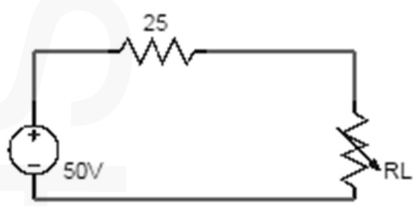
7. The parameters Y_{11}, Y_{12}, Y_{21}, Y_{22} are called?

a) Open circuit impedance parameters b) Short circuit admittance parameters c) Inverse transmission parameters d) Transmission parameters

8. Determine the resonant frequency for the specifications: R = 10Ω, L = 0.1H, C = 10μF

a) 158 b) 159 c) 160 d) 161

9. In the circuit shown determine the value of load resistance when the load resistance draws maximum power?



a) 25 b) 50 c) 75 d) 100

10. What happens to the voltage across the inductor when the Q factor decreases?

a) Increases b) Decreases c) Remains the same d) Becomes zero

11 The armature of DC motor is laminated to ____________

a) To reduce mass b) To reduce hysteresis loss c) To reduce eddy current loss d) To reduce inductance

12 What is the effect of demagnetizing component of armature reaction in a DC machine?

a) Reduces generator emf b) Increases armature speed c) Reduces interpole flux density d) Results in sparking trouble

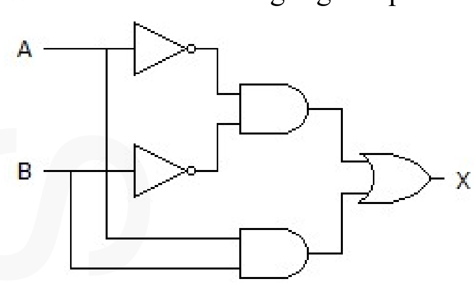
13 Direction of rotation of motor is determined by ____________

a) Faraday's law b) Lenz's law c) Coulomb's law d) Fleming's left-hand rule

14 External Characteristic of a DC machine is a plot of _____________

a) Generated emf and load current b) Terminal voltage and load current c) Generated voltage and field current d) Armature current and field current

15	Which losses can be identified from Swinburne's test?							
	a)	No-load core loss	b)	Windage and friction loss	c)	No-load and windage and friction loss	d)	Stray load loss
16	For a 20kVA transformer with a turn ratio of 0.4 ,what amount of total power is transferred inductively?							
	a)	10kVA	b)	8kVA	c)	50kVA	d)	12kVA
17	For a transformer with primary turns 100, secondary turns 400, if 200 V is applied at primary we will get ___________ at secondary							
	a)	3200 V	b)	1600 V	c)	800 V	d)	80V
18	The full-load copper loss of a transformer is 1600 W. At half-load, the copper loss will be							
	a)	1600 W	b)	6400 W	c)	400 W	d)	800 W
19	The efficiency of transformer compared to electric motor of the same power are ______							
	a)	About the same	b)	Much smaller	c)	Much higher	d)	Can't comment
20	A transformer having maximum efficiency at 75% full load will have ratio of iron loss and full load copper loss equal to							
	a)	4/3	b)	3/4	c)	9/16	d)	16/9
21	Which of the following is an example of a digital Electronic?							
	a)	Computers	b)	Information appliances	c)	Digital cameras	d)	All of the mentioned
22	What will be the output from a D flip-flop if D = 1 and the clock is low?							
	a)	No change	b)	Toggle between 0 and 1	c)	0	d)	1
23	The octal equivalent of the decimal number $(417)_{10}$ is ______							
	a)	$(641)_8$	b)	$(619)_8$	c)	$(640)_8$	d)	$(598)_8$
24	If A and B are the inputs of a half adder, the sum is given by ___________							
	a)	A AND B	b)	A OR B	c)	A XOR B	d)	A EX-NOR B
25	Convert binary to octal: $(110110001010)_2$ =							
	a)	$(5512)_8$	b)	$(6512)_8$	c)	$(4532)_8$	d)	$(6832)_8$
26	Which of the following logic expressions represents the logic diagram shown?							



	a)	X=AB'+A'B	b)	X=(AB)'+AB	c)	X=(AB)'+A'B'	d)	X=A'B'+AB

27 In S-R flip-flop, if Q = 0 the output is said to be ___________

a) Set b) Reset c) Previous state d) Current state

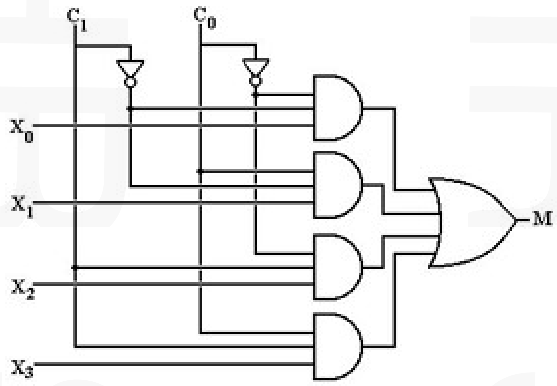
28 How is a J-K flip-flop made to toggle?

a) J = 0, K = 0 b) J = 1, K = 0 c) J = 0, K = 1 d) d) J = 1, K = 1

29 How many select lines would be required for an 8-line-to-1-line multiplexer?

a) 2 b) 4 c) 8 d) 3

30 In the given 4-to-1 multiplexer, if c1 = 0 and c0 = 1 then the output M is ___________



a) X_0 b) X_1 c) X_2 d) X_3

31 The area under the load curve represents

a) maximum demand b) load factor c) the average load on power system d) number of units generated

32 Which of the following is not a requirement for site selection of hydroelectric power plant?

a) Large catchment area b) Rocky land c) Sedimentation d) Availability of water

33 What is the effect of rise in temperature on sag when Ice and wind effect are eliminated?

a) Sag decreases b) Sag increases c) Sag remains constant d) Sag becomes zero

34 Why are the transmission lines transposed?

a) Reduce corona loss. b) Reduce skin effect. c) Prevent interference with the neighbouring telephone lines. d) Prevent short circuit between any two lines.

35 On which factor does the capacitance of the cable depend upon?

a) Length of cable. b) Relative permittivity of dielectric used in cable. c) Ratio of sheath diameter and core diameter. d) All of the above

36 Which type of distribution is preferred in residential areas?

a) Single phase, two wire. b) Three phase, three wire c) Three phase, four wire d) Two phase, four wire

37 The conductor carries more current on the surface in comparison to its core. What is this phenomenon called?

a) Corona b) Ferranti effect c) Skin effect d) Proximity effect

38 Critical disruptive voltage is a voltage at which

a) corona is just initiated b) corona is visible with blue or violet colour c) both a and b d) None of these

39 Transmission line constants are

a) Capacitance b) Inductance c) Resistance d) All of these

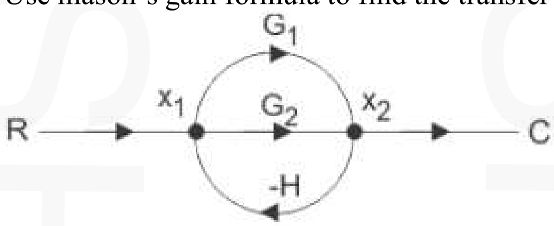
40 Voltage gradient on a transmission line conductor is highest

a) At the surface of the conductor b) At the centre of the conductor c) At the distance equal to one radius from the surface d) None of these

41 Transfer function of a system is defined as the ratio of output to input in

a) Z-transform b) Fourier transform c) Laplace transform d) All of these

42 Use mason's gain formula to find the transfer function of the given figure:



a) G_1+G_2 b) G_1+G_1/(1-G_1H+G_2H) c) G_1+G_2/(1-G_1H+G_2H) d) $G_1 - G_2$

43 Routh Hurwitz criterion gives

a) Number of roots in the right half of the s-plane b) Value of the roots c) Number of roots in the left half of the s-plane d) Number of roots in the top half of the s-plane

44

a) u(t) b) sgn(t) c) r(t) or r(n) d) None of the above

45 A system is said to be _____ if its response is dependent upon the present and past inputs and doesn't depend upon future input

a) Causal b) Non Causal c) Time variant d) None of the above

46 Laplace transform if sin(at)u(t) is

a) s/a^2+s^2 b) a/a^2+s^2 c) s^2/a^2+s^2 d) a^2+/a^2+s

47 Is the function y[n] = x[n-1] – x[n-56] causal?

a) The system is non causal b) The system is causal c) Both causal and non causal d) None of the mentioned

48 Find the Z-transform of a^n u(n);a>0.

a) z/z−a b) z/z+a c) c1/1−az d) 1/1+az

49 Which of the following is the process of 'aliasing'?

a) Peak overlapping b) Phase overlapping c) Amplitude overlapping d) Spectral overlapping

50 The capacitance, in force-current analogy, is analogous to

a) Momentum b) Velocity c) Displacement d) Mass

*****